

# Analysis group meeting

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Cooking for Pass0v8, started Friday 24 January.

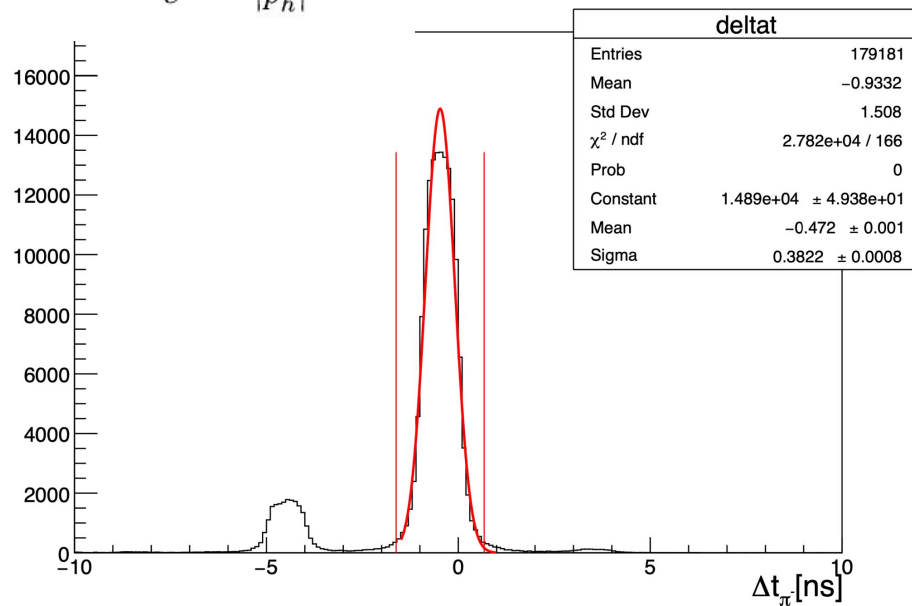
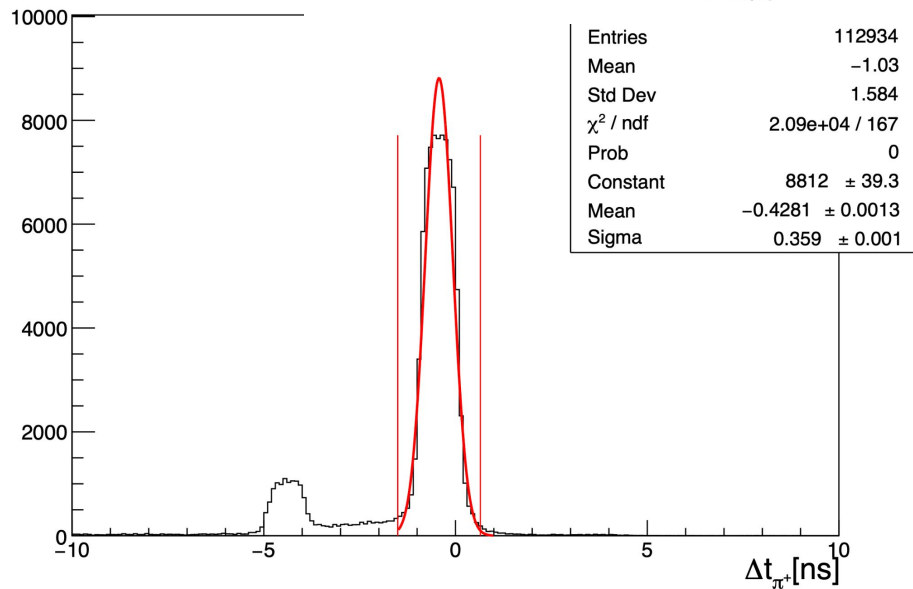
|                                        |        |       |      |  |
|----------------------------------------|--------|-------|------|--|
| rgd-drqa-pass0v8_018316-18316          | 88.89% | 45    | 5    |  |
| rgd-drqa-pass0v8_ShortEmpRand-18305x45 | 90.02% | 1913  | 191  |  |
| rgd-drqa-pass0v8_CxC-18339x125         | 94.49% | 5625  | 310  |  |
| rgd-drqa-pass0v8_LD2-18307x147         | 92.03% | 6615  | 527  |  |
| rgd-drqa-pass0v8_CuSn-18347x301        | 92.09% | 13545 | 1072 |  |

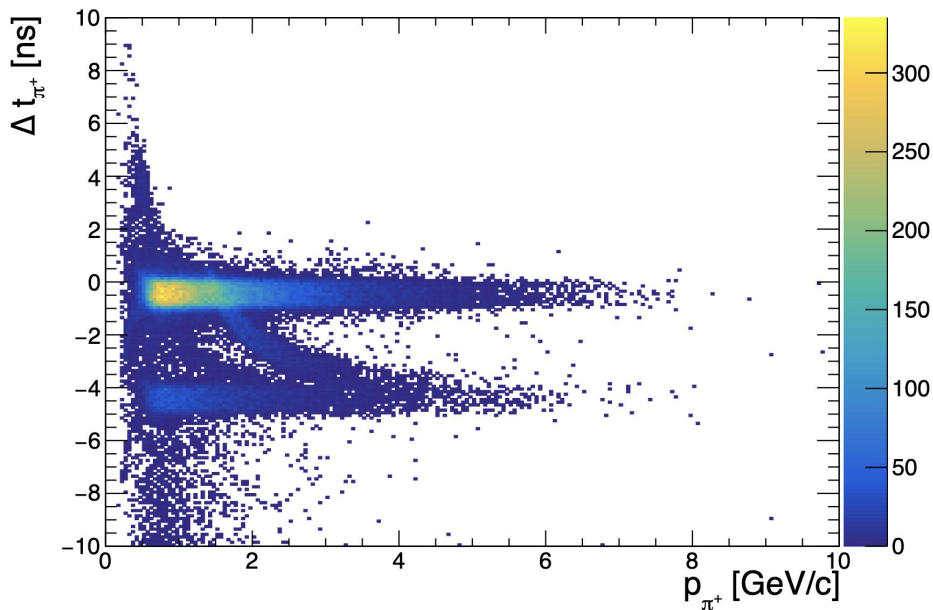
# Vertex study: effect of a cut on $\Delta t$ on $V_z$

Use Pass0v7 data, run 018563, 018564, and 018565, **Cu** target

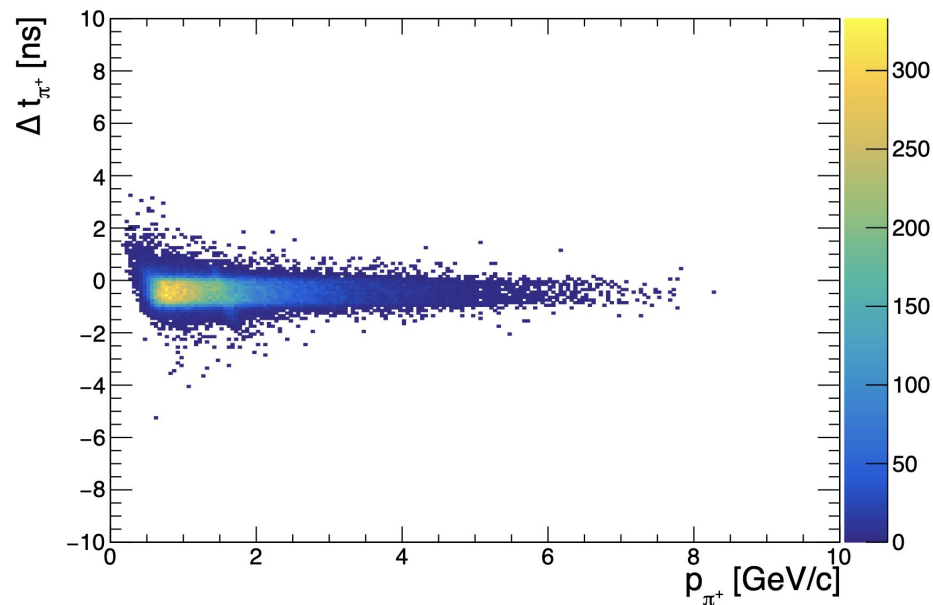
Cut electron: pid=11,status<0,-5<chi2pid<5 and  $-8.477 - 3 * 0.675 < V_z < -8.477 + 3 * 0.675$  cm

$$\Delta t = t_{\text{FTOF}} - t_{\text{start time}} - \frac{l}{c\beta_{h(p)}} = t_{\text{FTOF}} - t_{\text{start time}} - \frac{l}{c} \frac{\sqrt{|\vec{p}_h|^2 + m_h^2}}{|\vec{p}_h|}$$

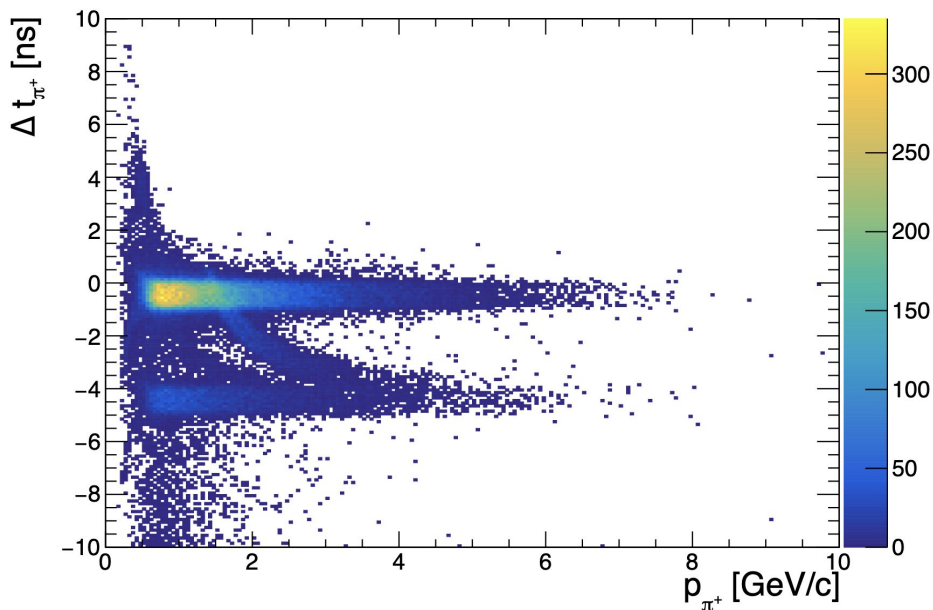




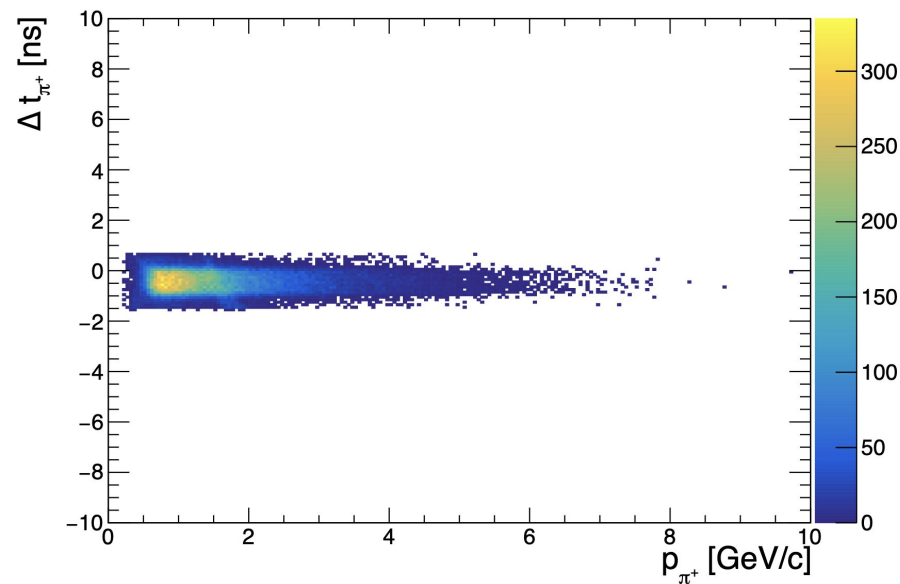
pid = 211



pid = 211 and  $-10 < \chi^2_{\text{pid}} < 10$

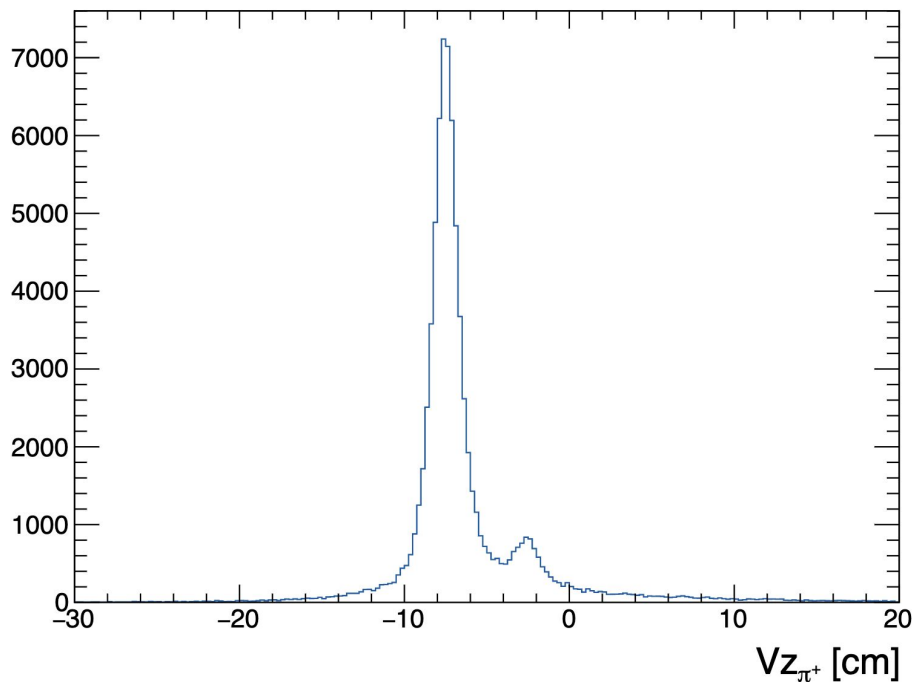


pid = 211

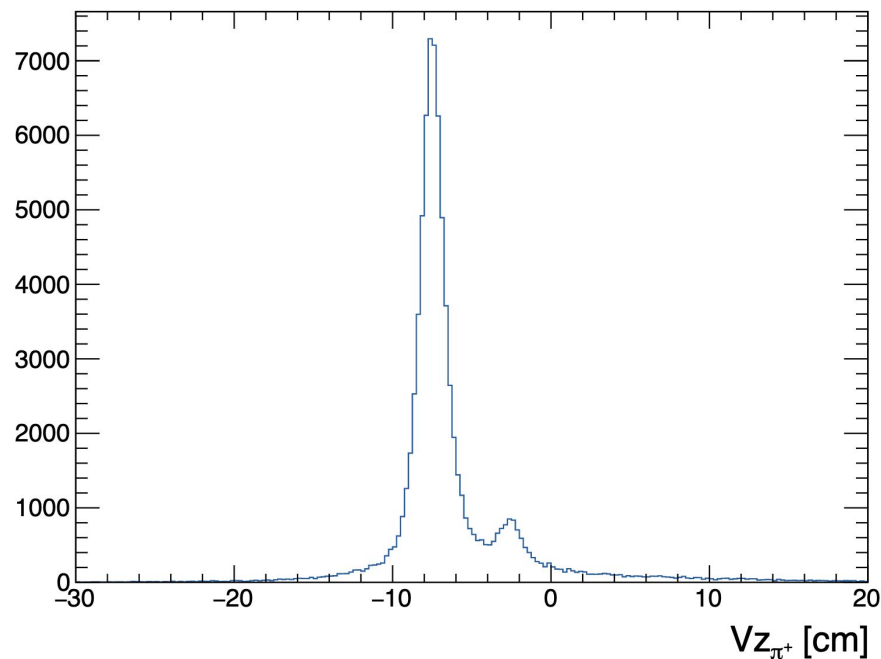


pid = 211

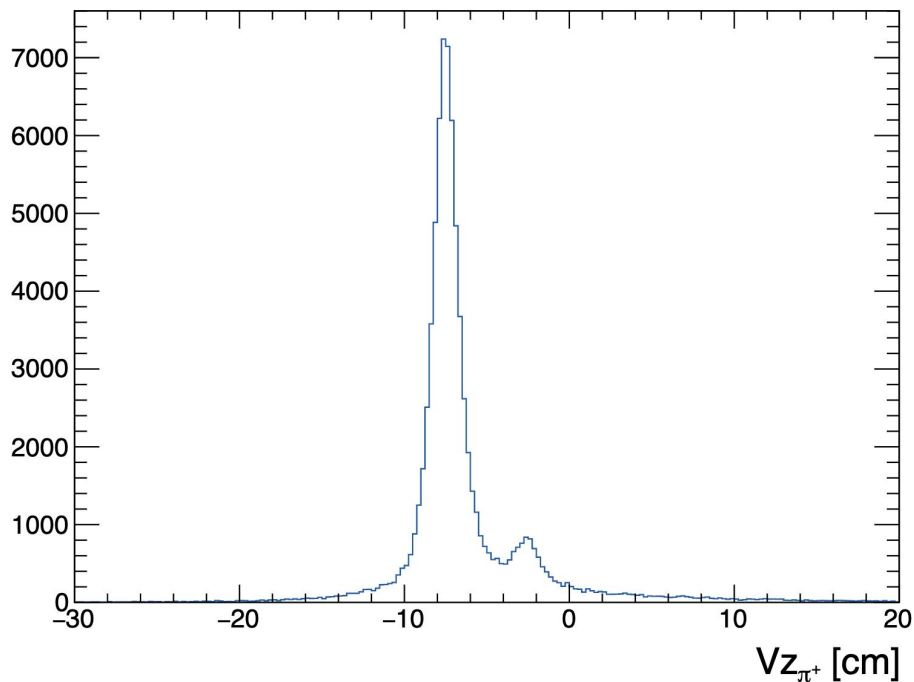
$$-0.4281 - 3 \cdot 0.359 < \Delta t < -0.4281 + 3 \cdot 0.359$$



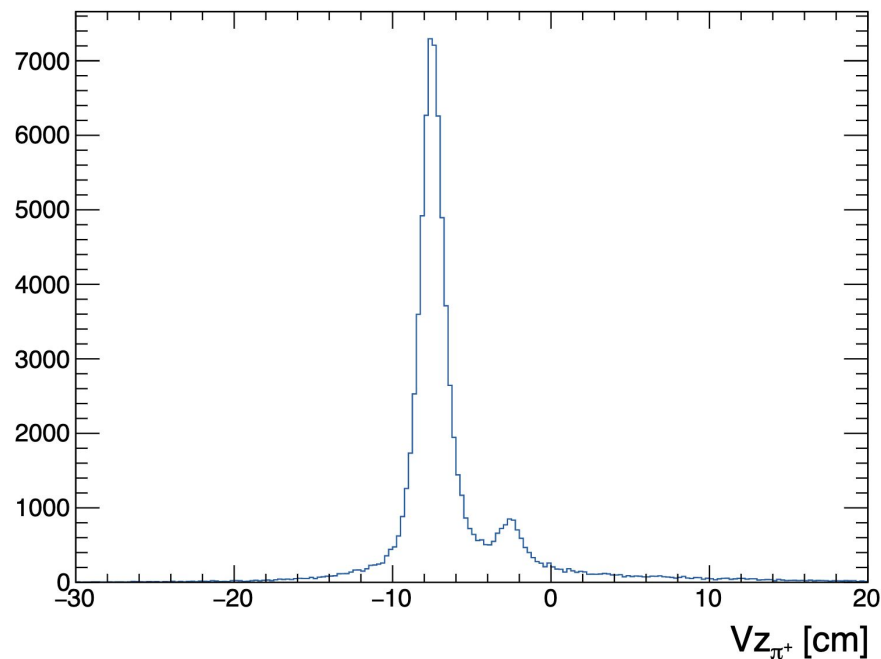
$\text{pid} = 211$  and  $-10 < \chi^2_{\text{pid}} < 10$



$\text{pid} = 211$   
 $-0.4281 - 3 \cdot 0.359 < \Delta t < -0.4281 + 3 \cdot 0.359$



$\text{pid} = 211$  and  $-10 < \chi^2_{\text{pid}} < 10$



$\text{pid} = 211$   
 $-0.4281 - 3 \cdot 0.359 < \Delta t < -0.4281 + 3 \cdot 0.359$

- A cut on  $\Delta t$  doesn't reduce the secondary peak in  $V_z$ .
- Started to develop a momentum depend on  $\Delta t$  [bin size: 0.5 GeV/c]
- Try to look to other variables if I select the secondary peak (cut on  $V_z > -4$  cm), I haven't seen any dependence

- Have the run conditions (for each target, magnetic field and current):
  - LD2 configuration
    - magnetic field : solenoid -1, torus 1/-1
    - current: 45,50,60
  - CxC configuration
    - magnetic field : solenoid -1, torus 1/-1
    - current: 50,75
  - CuSn configuration
    - magnetic field : solenoid -1, torus 1/-1
    - current 90,95,100,110,130,150

# Background merging

- Have the run conditions (for each target, magnetic field and current)
- Get the decoded files, I can use the Pass0v8 files
- Produce between 20k to 100k background event per run condition:
  - `trigger-filter -b [trigger bit (0-63)] -o [output file] input-file-list -s (switch from Faraday cup current to EPIC) -c (threshold current value)`
  - trigger bit for the Faraday cup `0x0000008000000000`
  - threshold current value: value - 5nA
  - Since Faraday cups had issue, I can use the EPIC bank to get the current
- For the review one target with good luminosity runs should be enough (CuSn?)

# RG-D study: $\pi^0$ analysis

Nothings to show.

I started working on the event mixing method, based on Neutral Pion Multiplicity Ratios from SIDIS Lepton-nuclear Scattering T.Mineeva, W.Brooks, K.Joo

Need to be able to process next and previous event with the HIPO C++ library, I had to develop code to do that. And also be able to come back to a specific event.

```
Event number: 1001
Event number: 1002
Event number: 1001
Event number: 1000
Event number: 999
Event number: 998
Event number: 997
Event number: 996
Event number: 995
Event number: 994
Event number: 993
Event number: 992
Event number: 1001
```

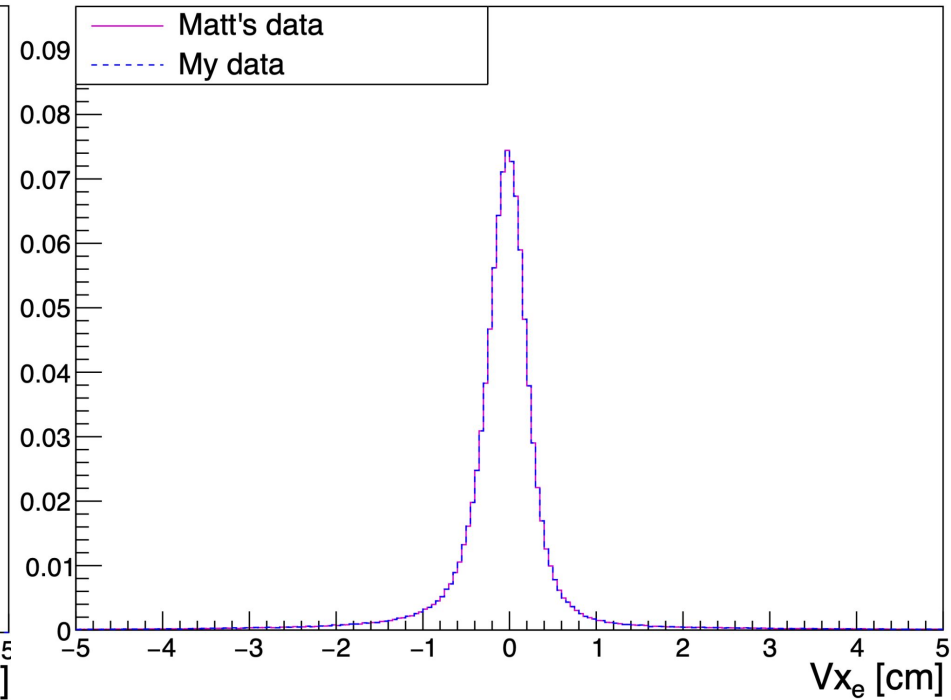
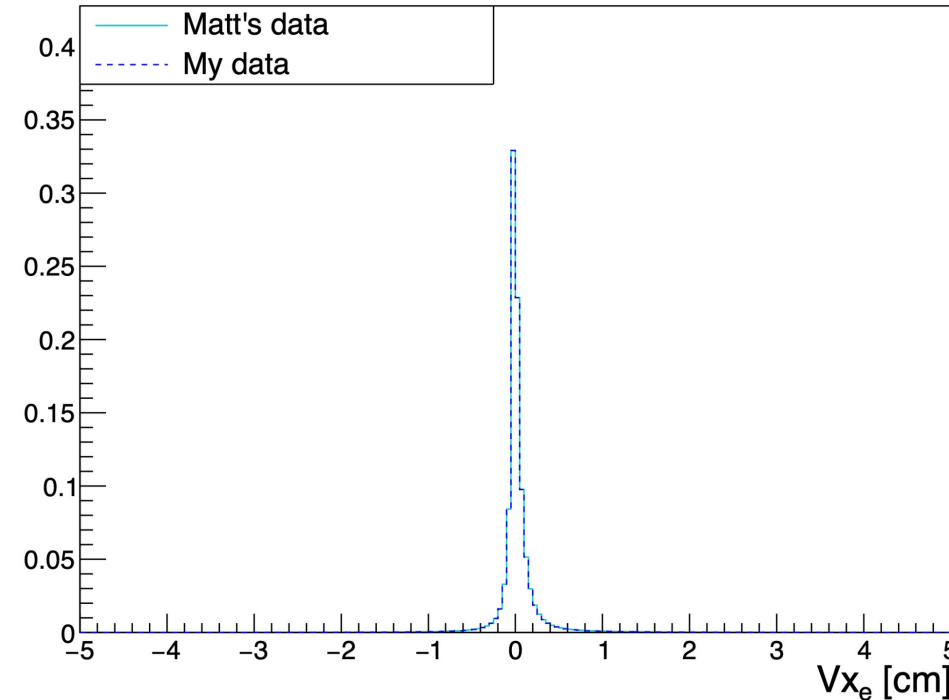
- RG-D analysis:
  - Cross check with Matt the  $V_x$ ,  $V_y$  and  $V_z$  mean and sigma for CxC
  - Cross check for the nuclear transparency vs  $Q^2$
  - Work on  $\pi^0$  analysis:
    - Improve  $\pi^0$  opening angle between the emitting scattered electron and the photon as function of the momentum ([Analysis note](#) by Stefan Diehl, Andrey Kim, Kyungseon Joo)
    - Study the  $E_y$  cut effect on the two photons mass
    - Implement the event mixing method
- Background merging, generate 100k events for one configuration
- AI for ALERT:
  - Redo the efficiency/purity analysis using luminosity simulation to have a better background
  - Comparison between conventional and AI track finding

# Backup slides

Use Pass0v7 data, run 018563, 018564, and 018565, **Cu** target

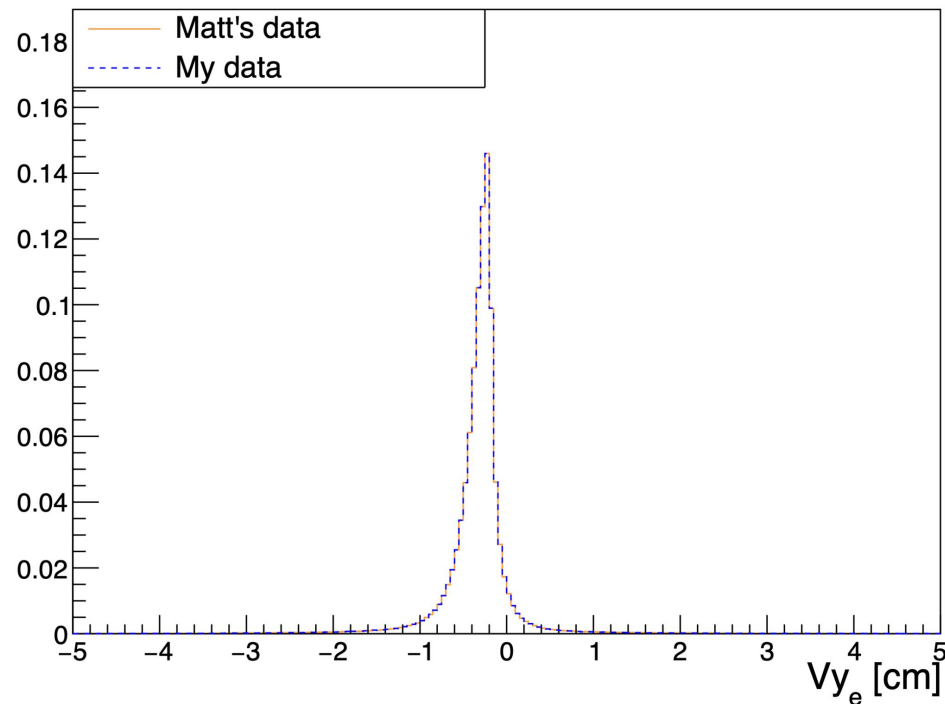
Electron, sector 1

Electron, sector 6

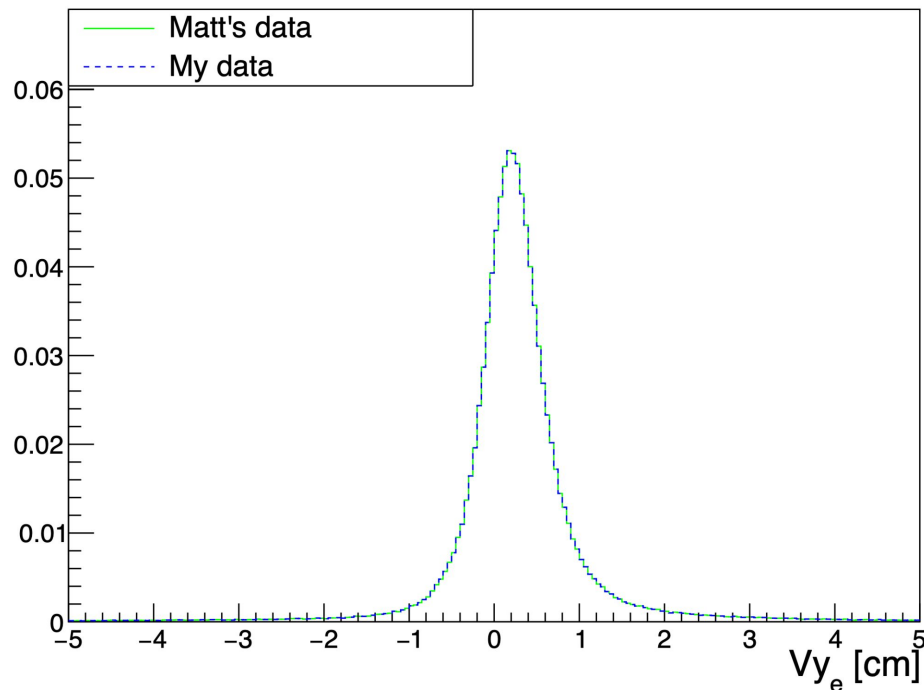


Use Pass0v7 data, run 018563, 018564, and 018565, **Cu** target

Electron, sector 2



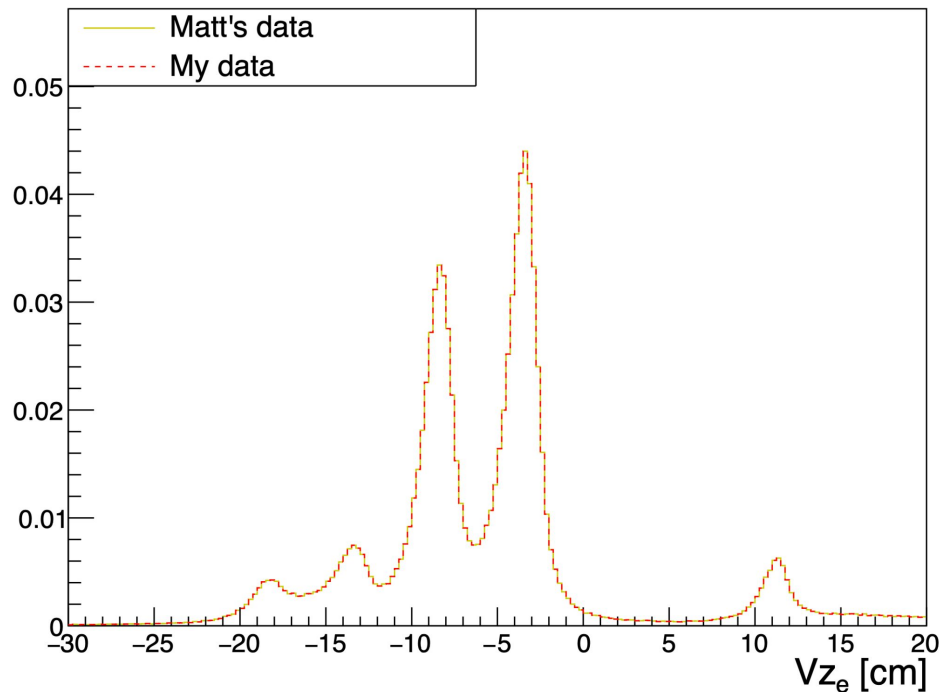
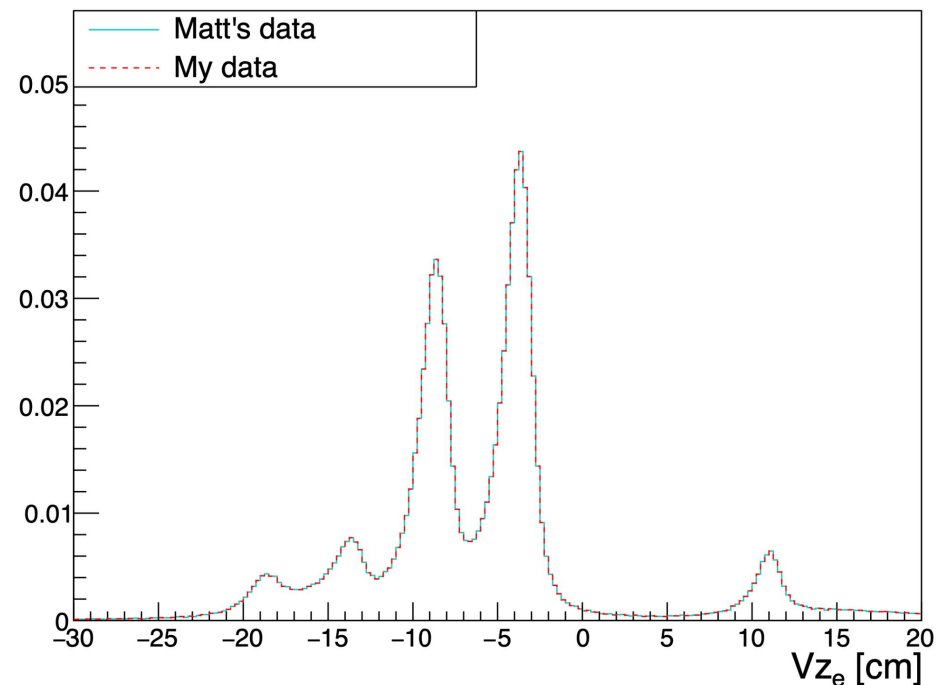
Electron, sector 4



Use Pass0v7 data, run 018563, 018564, and 018565, **Cu** target

Electron, sector 1

Electron, sector 3



# Vertex study: Copper, Electron

| Electron | V <sub>x</sub> |               | V <sub>y</sub> |               |
|----------|----------------|---------------|----------------|---------------|
|          | $\mu$ [cm]     | $\sigma$ [cm] | $\mu$ [cm]     | $\sigma$ [cm] |
| Sector 1 | 0.001142       | 0.057957      | -0.309368      | 0.258175      |
| Sector 2 | 0.21525        | 0.342234      | -0.290134      | 0.135542      |
| Sector 3 | 0.051030       | 0.222614      | -0.155606      | 0.161925      |
| Sector 4 | -0.025516      | 0.098360      | -1.910965      | 7.163615      |
| Sector 5 | -0.259626      | 0.250884      | -0.062024      | 0.137291      |
| Sector 6 | -0.028364      | 0.216857      | -0.187878      | 0.161606      |

| Electron                       | V <sub>z</sub> |               |
|--------------------------------|----------------|---------------|
|                                | $\mu$ [cm]     | $\sigma$ [cm] |
| Entrance of the empty LD2 cell | -18.437182     | 0.840693      |
| Exit of the empty LD2 cell     | -13.688906     | 0.738610      |
| first foil                     | -8.477611      | 0.674902      |
| second foil                    | -3.620864      | 0.795920      |

# Vertex study: Copper, $\pi^+$

| $\pi^+$  | V <sub>x</sub> |               | V <sub>y</sub> |               |
|----------|----------------|---------------|----------------|---------------|
|          | $\mu$ [cm]     | $\sigma$ [cm] | $\mu$ [cm]     | $\sigma$ [cm] |
| Sector 1 | -0.030302      | 0.147119      | -0.462450      | 0.423933      |
| Sector 2 | 0.136321       | 0.361236      | -0.232646      | 0.176071      |
| Sector 3 | -0.080255      | 0.746294      | -0.258453      | 0.124568      |
| Sector 4 | 0.025255       | 0.147954      | 0.112324       | 0.450481      |
| Sector 5 | -0.106297      | 0.325653      | -0.143978      | 0.142895      |
| Sector 6 | 0.034948       | 0.436025      | -0.124890      | 0.123717      |

| $\pi^+$     | V <sub>z</sub> |               |
|-------------|----------------|---------------|
|             | $\mu$ [cm]     | $\sigma$ [cm] |
| first foil  | -7.463966      | 0.765427      |
| second foil | -2.645098      | 0.594126      |

# Vertex study: Copper, $\pi^-$

| $\pi^-$  | $V_x$      |               | $V_y$      |               |
|----------|------------|---------------|------------|---------------|
|          | $\mu$ [cm] | $\sigma$ [cm] | $\mu$ [cm] | $\sigma$ [cm] |
| Sector 1 | 0.017426   | 0.144236      | -0.218028  | 0.484644      |
| Sector 2 | 0.375652   | -0.387013     | -0.302328  | 0.146333      |
| Sector 3 | 0.180487   | 0.486722      | -0.121995  | 0.137190      |
| Sector 4 | -0.061903  | 0.180403      | 0.247462   | 0.739599      |
| Sector 5 | -0.371188  | 0.308023      | -0.055174  | 0.134398      |
| Sector 6 | -0.079255  | 0.327967      | -0.209298  | 0.153217      |

| $\pi^-$     | $V_z$      |               |
|-------------|------------|---------------|
|             | $\mu$ [cm] | $\sigma$ [cm] |
| first foil  | -8.394546  | 0.853582      |
| second foil | -3.325793  | 0.630868      |

# Vertex study: Tin

| $\pi^+$  | V <sub>x</sub> |               | V <sub>y</sub> |               |
|----------|----------------|---------------|----------------|---------------|
|          | $\mu$ [cm]     | $\sigma$ [cm] | $\mu$ [cm]     | $\sigma$ [cm] |
| Sector 1 | -0.030258      | 0.139951      | -0.565634      | 1.335854      |
| Sector 2 | 0.145823       | 0.386057      | -0.226411      | 0.158705      |
| Sector 3 | -0.070658      | 0.540834      | -0.255098      | 0.124184      |
| Sector 4 | 0.023502       | 0.139429      | 0.073561       | 0.904049      |
| Sector 5 | -0.114724      | 0.321049      | -0.144784      | 0.135346      |
| Sector 6 | 0.025608       | 0.406643      | -0.128251      | 0.112621      |

| $\pi^+$     | V <sub>z</sub> |               |
|-------------|----------------|---------------|
|             | $\mu$ [cm]     | $\sigma$ [cm] |
| first foil  | -5.897985      | 0.642928      |
| second foil | -2.589883      | 0.683810      |

# Vertex study: $\pi^-$

| $\pi^-$  | $V_x$      |               | $V_y$      |               |
|----------|------------|---------------|------------|---------------|
|          | $\mu$ [cm] | $\sigma$ [cm] | $\mu$ [cm] | $\sigma$ [cm] |
| Sector 1 | 0.018797   | 0.141719      | -0.126843  | 0.664250      |
| Sector 2 | -3.603485  | -5.011191     | -0.299321  | 0.140911      |
| Sector 3 | 0.168048   | 0.439710      | -0.114308  | 0.153320      |
| Sector 4 | -0.062215  | 0.187322      | 0.293006   | -1.120913     |
| Sector 5 | -0.376369  | 0.316821      | -0.061996  | 0.127000      |
| Sector 6 | -0.067269  | 0.328346      | -0.206470  | 0.160979      |

| $\pi^-$     | $V_z$      |               |
|-------------|------------|---------------|
|             | $\mu$ [cm] | $\sigma$ [cm] |
| first foil  | -4.251155  | 1.409331      |
| second foil | -3.311006  | 0.717284      |

| Electron    | V <sub>x</sub> |               | V <sub>y</sub> |               |
|-------------|----------------|---------------|----------------|---------------|
|             | $\mu$ [cm]     | $\sigma$ [cm] | $\mu$ [cm]     | $\sigma$ [cm] |
| All sectors | -0.003715      | 0.138771      | -0.178137      | 0.179863      |
| Sector 1    | 0.002255       | 0.054381      | -0.313036      | 0.237254      |
| Sector 2    | 0.266186       | 0.296362      | -0.291218      | 0.125726      |
| Sector 3    | 0.058212       | 0.206895      | -0.149784      | 0.150069      |
| Sector 4    | -0.027312      | 0.096397      | 0.227775       | 7.505927      |
| Sector 5    | -0.259326      | 0.238517      | -0.063601      | 0.132235      |
| Sector 6    | -0.032030      | 0.203369      | -0.190754      | 0.152068      |

| Electron                       | V <sub>z</sub> |               |
|--------------------------------|----------------|---------------|
|                                | $\mu$ [cm]     | $\sigma$ [cm] |
| Entrance of the empty LD2 cell | -18.425990     | -1.227214     |
| Exit of the empty LD2 cell     | -13.878048     | 1.071416      |
| first foil                     | -8.524542      | 0.679598      |
| second foil                    | -3.638532      | 0.811611      |